Practice workbook · Explain diffusion across a partially permeable membrane in dialysis and Visking tubing

How dialysis fluid removes urea but keeps glucose and salts, why glucose is added, and Visking tubing models where small molecules pass and starch stays.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-PARTIALLY-PERMEABLE-MEMBRANE-DIALYSIS

# 1. Question 8 · part b.i · 4 marks

January 2020 Paper 1

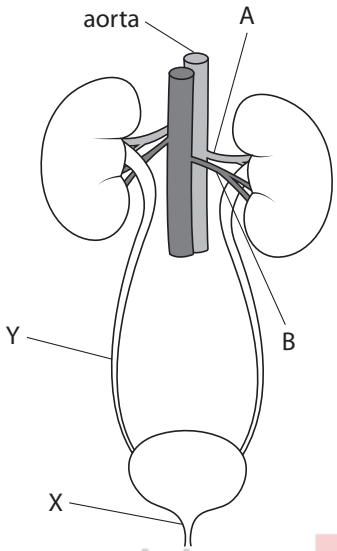
Worked-example retry · Attempt part b.i only.

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**8** The diagram shows part of the human renal system.




(a) (i) Give the names of structures X and Y.

**(2)**

structure X .....

structure Y .....

(ii) Describe the function of the aorta.

**(2)**

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(iii) Compare the composition of the blood in vessels A and B.

**(2)**

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# 1. Question 8 · part b.i · 4 marks

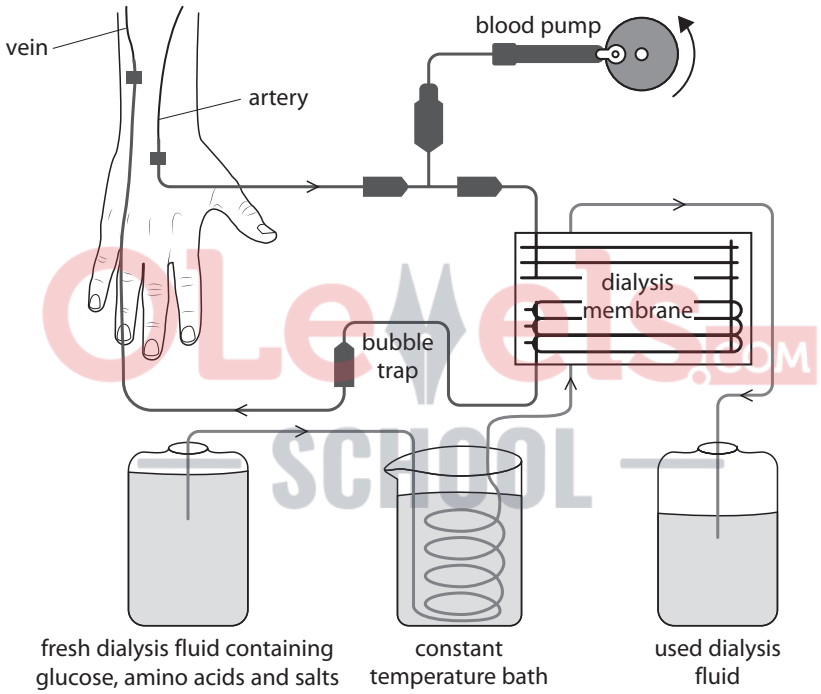
January 2020 Paper 1

Worked-example retry · Attempt part b.i only.

(b) Dialysis is a treatment used for people with kidney failure.

Blood containing waste substances is pumped through a dialysis machine. The dialysis machine filters the waste from the blood. The cleaned blood is then returned to the patient.

The diagram shows how a dialysis machine filters the blood of a person with kidney failure.




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# 1. Question 8 · part b.i · 4 marks

January 2020 Paper 1

Worked-example retry · Attempt part b.i only.

(i) Explain how diffusion removes waste substances from the patient's blood inside the dialysis machine.

**(4)**

(ii) A chemical is added to the patient's blood as it enters the dialysis machine. This chemical prevents the blood from clotting.

Explain why it is important to add this chemical as the blood enters the machine, rather than just before the blood returns to the body.

**(3)**

**(Total for Question 8 = 13 marks)**

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# 1. Question 8 · part b.i · 4 marks · Working and answer

Working and answer

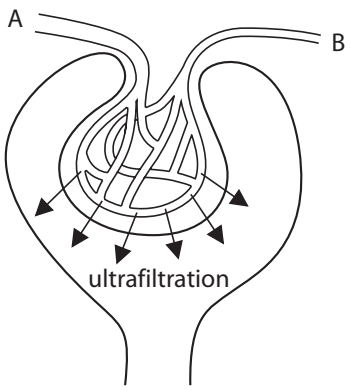
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## 2. Question 6 · part b · 3 marks

June 2021 Paper 1

Further practice · Attempt part b only.

**6** (a) The diagram shows a Bowman's capsule and its associated blood supply.




(i) Describe what is meant by the term **ultrafiltration**.

**(2)**

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(ii) State three differences between the composition of blood in vessel A and the composition of blood in vessel B.

**(3)**

1 .....

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2 .....

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3 .....

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## 2. Question 6 · part b · 3 marks

June 2021 Paper 1

Further practice · Attempt part b only.

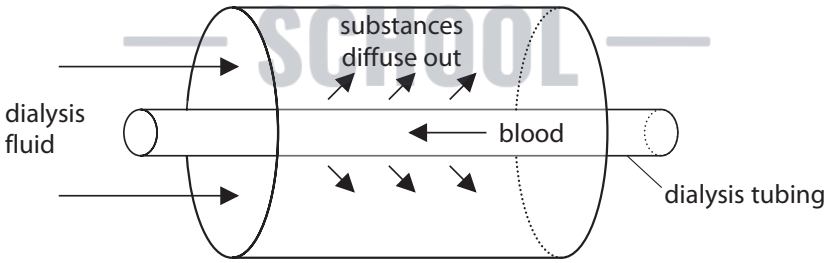
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(iii) Explain how blood vessel A and blood vessel B help the process of ultrafiltration.

**(4)**

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(b) The diagram shows part of a kidney dialysis machine.




The dialysis fluid contains glucose at the same concentration as in the blood flowing through the dialysis tubing.

Suggest the effect on a person undergoing dialysis if the concentration of glucose was below that of the blood flowing through the dialysis tubing.

**(3)**

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**(Total for Question 6 = 12 marks)**

## 2. Question 6 · part b · 3 marks · Working and answer

Working and answer

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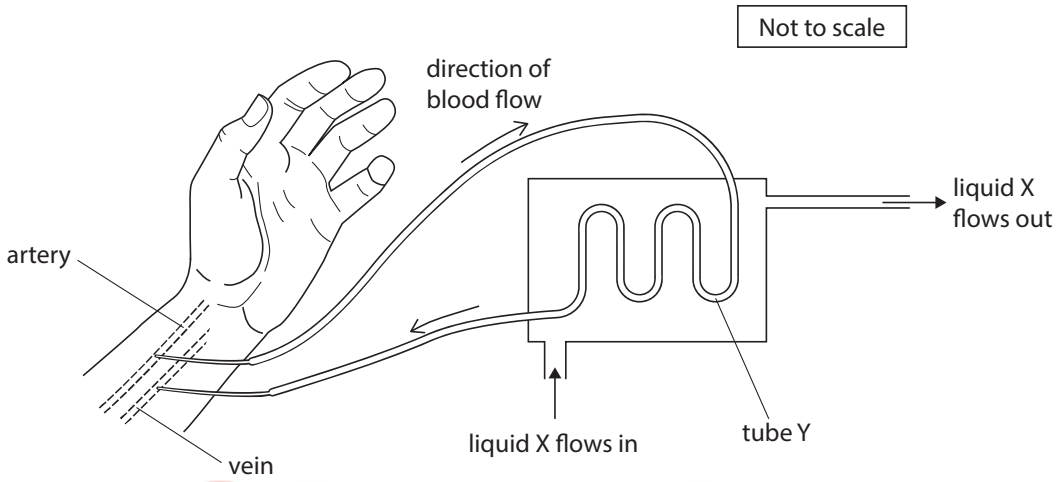
# 3. Question 5 · part b · 3 marks

June 2025 Paper 1R

Further practice · Attempt part b only.

**5** The diagram shows a kidney dialysis machine.

The wall of tube Y is partially permeable so it allows the passage of small molecules.




(a) (i) Explain why the blood flow is in the direction shown in the diagram.

**(4)**

(ii) State which part of the kidney nephron has the same role as tube Y.

**(1)**

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## 3. Question 5 · part b · 3 marks

June 2025 Paper 1R

Further practice · Attempt part b only.

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(iii) Explain why tube Y has a length of several metres in the dialysis machine.

**(2)**

(iv) State one difference between the composition of liquid X as it flows into the machine and as it flows out of the machine.

**(1)**

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(b) Glucose is added to liquid X before it passes into the dialysis machine.

Explain why glucose is added.

**(3)**

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## 3. Question 5 · part b · 3 marks

June 2025 Paper 1R

Further practice · Attempt part b only.

(c) Explain why people who have to use kidney dialysis machines are often told to eat less protein in their diet.

**(4)**

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**(Total for Question 5 = 15 marks)**

Working and answer

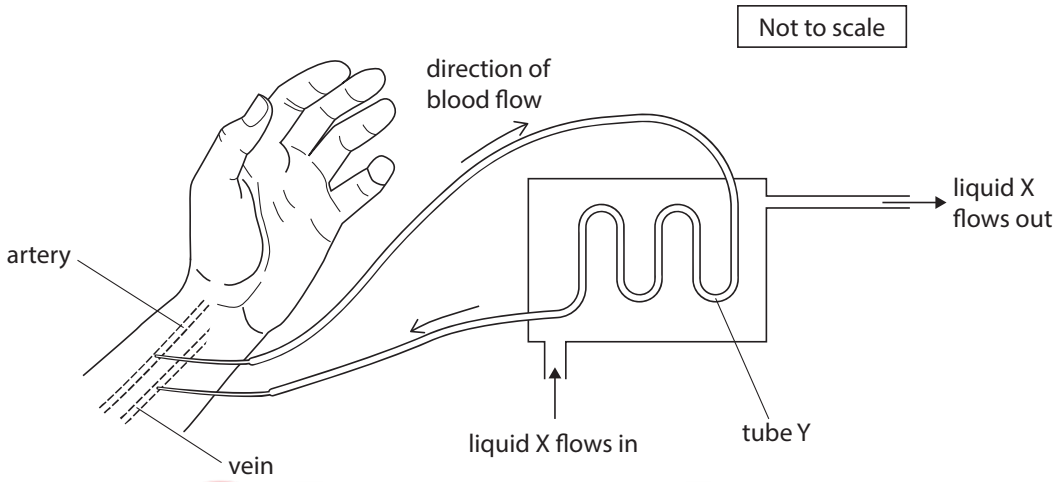
# 4. Question 5 · part a.iv · 1 mark

June 2025 Paper 1R

Further practice · Attempt part a.iv only.

**5** The diagram shows a kidney dialysis machine.

The wall of tube Y is partially permeable so it allows the passage of small molecules.




(a) (i) Explain why the blood flow is in the direction shown in the diagram.

**(4)**

(ii) State which part of the kidney nephron has the same role as tube Y.

**(1)**

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# 4. Question 5 · part a.iv · 1 mark

June 2025 Paper 1R

Further practice · Attempt part a.iv only.

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(iii) Explain why tube Y has a length of several metres in the dialysis machine.

**(2)**

(iv) State one difference between the composition of liquid X as it flows into the machine and as it flows out of the machine.

**(1)**

(b) Glucose is added to liquid X before it passes into the dialysis machine.

Explain why glucose is added.

**(3)**

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## 4. Question 5 · part a.iv · 1 mark · Working and answer

Working and answer

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# 5. Question 2 · part b · 5 marks

January 2022 Paper 1

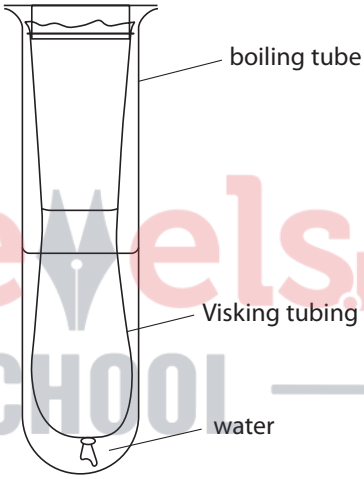
Further practice · Attempt part b only.

**2** A student investigates absorption in the ileum by using a model gut made of Visking tubing.

This is the student's method.

- tie a knot in one end of a 15 cm length of Visking tubing to make a bag
- place 5 $cm^3$ of starch suspension and 5 $cm^3$ of glucose solution into the bag
- rinse the outside of the bag with water
- place the bag into a boiling tube containing distilled water
- leave the bag for one hour

The diagram shows the student's apparatus.




After an hour the contents of the Visking tubing bag are tested for glucose and starch.

The water in the boiling tube is also tested for glucose and starch.

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# 5. Question 2 · part b · 5 marks

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Further practice · Attempt part b only.

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(a) (i) Describe how the student should test the contents of the Visking tubing bag for glucose.

**(4)**

(ii) Describe how the student should test the contents of the Visking tubing bag for starch.

**(3)**

(iii) Explain why the outside of the Visking tubing bag was rinsed with water before being placed in the boiling tube.

**(3)**

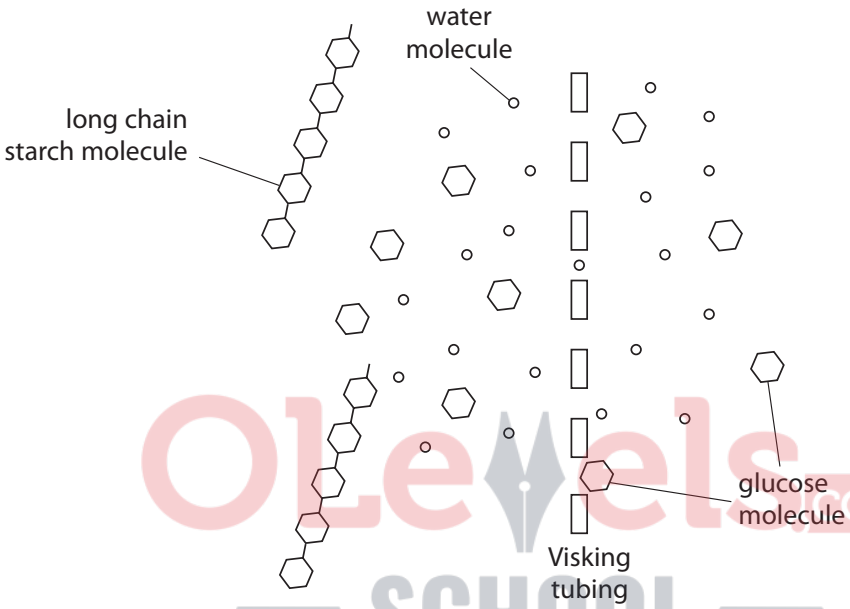
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# 5. Question 2 · part b · 5 marks

January 2022 Paper 1

Further practice · Attempt part b only.

(b) The student draws this diagram to explain the results obtained.




Explain the results the student obtains.

Refer to the diagram in your answer.

**(5)**

**(Total for Question 2 = 15 marks)**

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## 5. Question 2 · part b · 5 marks · Working and answer

Working and answer

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# 6. Question 5 · part b.iii · 3 marks

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

**5** Glucose is a product of digestion.

Glucose moves from the small intestine into the blood by diffusion.

(a) Explain why the rate of diffusion of glucose into the blood stream is affected by the concentration of glucose in the small intestine.

**(3)**

(b) A student uses different concentrations of glucose solution to investigate the rate of diffusion from the small intestine into the blood.

The list gives some of the equipment that the student uses in the investigation.

- Visking tubing (to represent the small intestine)
- boiling tube
- different concentrations of glucose solutions
- stop clock

(i) Name the chemical the student uses to test for glucose.

**(1)**

(ii) Name one other piece of equipment that the student would need to use to test for glucose.

**(1)**

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## 6. Question 5 · part b.iii · 3 marks

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

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(iii) Draw a diagram to show how the student could use the equipment to investigate the effect of glucose concentration on the rate of diffusion.

You should add these labels to your diagram.

- Visking tubing
- boiling tube
- sugar solution
- water

**(3)**

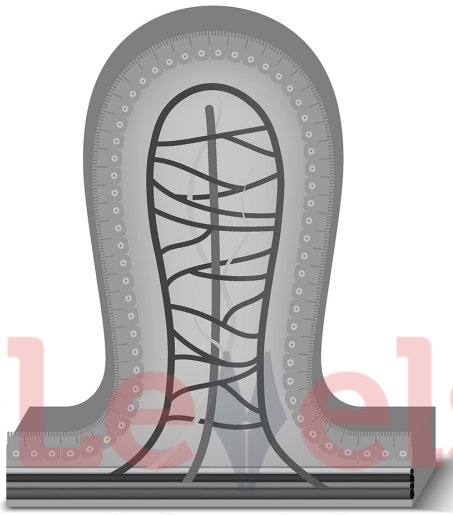
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# 6. Question 5 · part b.iii · 3 marks

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

(c) The diagram shows a structure found in the small intestine. This structure increases the rate of diffusion of molecules such as glucose from the small intestine into the blood stream.



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(i) Name this structure.

**(1)**

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(ii) Explain how these structures increase the rate of diffusion of glucose from the small intestine into the blood stream.

**(3)**

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**(Total for Question 5 = 12 marks)**

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## 6. Question 5 · part b.iii · 3 marks · Working and answer

Working and answer

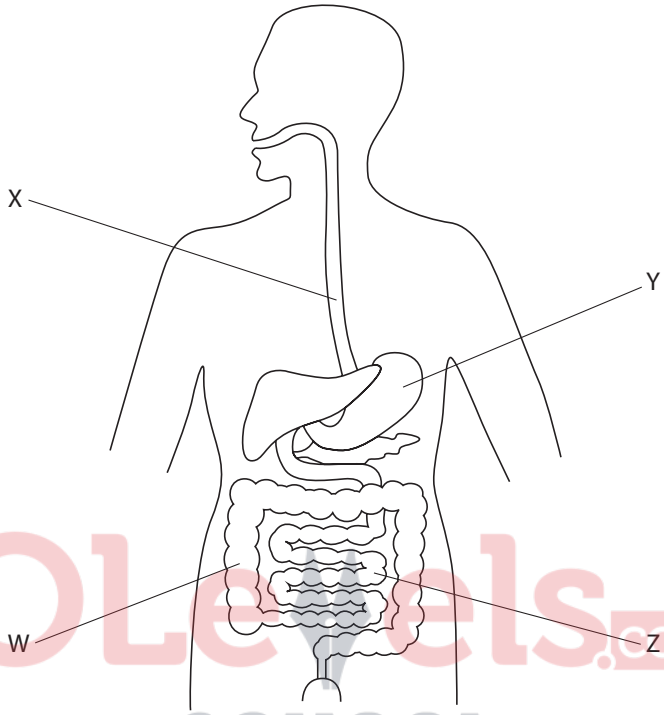
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# 7. Question 2 · part b.ii · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.ii only.

**2** (a) The diagram shows the human digestive system.




(i) Name the structures labelled W, X, Y and Z.

**(4)**

W .....

X .....

Y .....

Z .....

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# 7. Question 2 · part b.ii · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.ii only.

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(ii) The box gives words used to describe how food is moved through structure X towards structure Y.

| | | | |
|---|---|---|---|
| relax | coordination | contract | liquid |
| peristalsis | muscles | enzymes | bolus |

Use words from the box to complete the passage.

**(4)**

Food forms a ..... at the back of the mouth.

The food is swallowed and passes into structure X. The walls of structure X contain .....

which ..... to push the food towards structure Y.

This process is known as ..... .

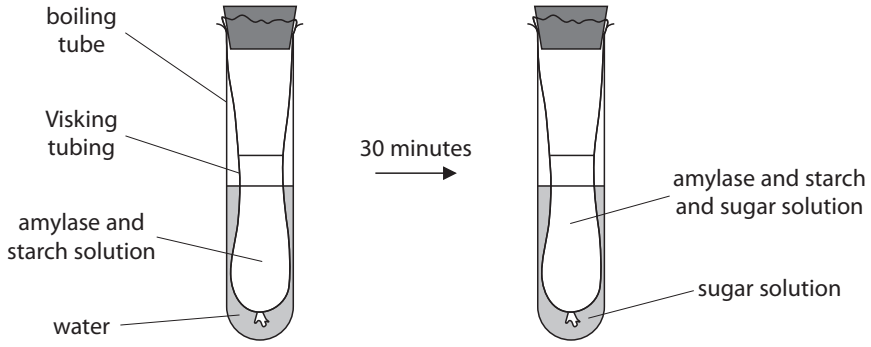
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# 7. Question 2 · part b.ii · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.ii only.

(b) The diagram shows some apparatus used to investigate the effect of an enzyme called amylase on starch.




(i) Name one area in the digestion system where amylase is produced.

(1)

(ii) Name the structure in the digestive system that the Visking tubing represents.

(1)

(iii) Explain the results of the investigation after 30 minutes.

(2)

(iv) Name the chemical used to test for sugar in the water surrounding the Visking tubing after 30 minutes.

(1)

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## 7. Question 2 · part b.ii · 1 mark · Working and answer

Working and answer

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